A local solution at every place need not be global

$$27u^6 + 3125v^{10} = 128w^{14}$$

Squarefree coefficients (a,b,c) = (3,5,2)

at 2

$3 + 5 \equiv 0 \pmod 8$

at 3

$10 \equiv 1 \pmod 3$

at 5

$6 \equiv 1 \pmod 5$

Local root variables may differ at each prime. Exact integer equality is unproved.